

Ruturaj Dilip Gawade

About Me

I build industrial automation that is reliable in real production, not only on paper.

My Story

During my diploma studies, I attended a workshop where I saw something that immediately clicked for me. On a screen, there was a ladder-like diagram, and that diagram was controlling real robots. It looked simple, but the impact was huge. I still remember thinking, if a few lines of logic can move a robot, then how cool will it be to see it controlling big machines in a factory.

That moment pushed me to take a course in automation, and soon after I started working in industry. In the beginning I was learning by doing, understanding sensors, electrical wiring, signals, and why machines fail when one small condition is wrong. Step by step, projects became bigger, and the problems became more real.

Within two years, I reached a point where I was not only writing the logic, but also guiding other engineers working under and alongside me. We worked shoulder to shoulder, programming systems, testing them, and commissioning complex automation for major customers. That journey shaped how I work today: structured, calm under pressure, and always focused on what will actually work on the shopfloor.

What I Do Today

I have 3 years of experience in industrial automation and commissioning, working across sectors such as automotive manufacturing, intralogistics and material handling, food processing, pharma production, and utility and process industries. My work covers planning, development, site commissioning, troubleshooting, customer coordination, and handover documentation, so systems stay stable not only during trials but also after go-live.

After delivering projects for customers across different countries and working in fast-paced shopfloor environments, I moved to Germany to pursue my master's in digital engineering. I enjoy combining real industrial experience with modern digital skills, and I am building my future around smarter, more connected automation systems.

Experience

Automation & Commissioning Engineer – Promatics Solutions

March 2021 – April 2024 | Pune, India

I worked on industrial automation projects from early customer discussions to on-site commissioning and after-sales support. My day-to-day goal was simple: make the system run reliably on the shopfloor and keep troubleshooting fast when production could not wait.

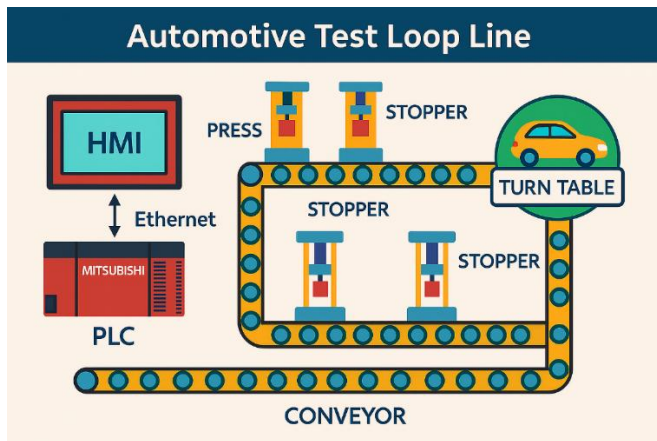
At Promatics Solutions, I handled both development work and real field execution. That meant writing logic, validating signals, checking wiring, coordinating with teams, and staying calm during critical restarts when the customer needed the line running again quickly.

- Led requirement and kickoff discussions with customers, clarified scope, I/O points, safety expectations, and acceptance criteria before development started.
- Developed PLC logic, HMI screens, and SCADA monitoring for complete automation systems, focusing on clean sequencing, safe interlocks, meaningful alarms, and operator-friendly recovery flows.
- Executed on-site commissioning and system integration, including loop checks, signal validation, sensor calibration, dry runs, hot trials, and final site acceptance testing.
- Managed industrial communication and troubleshooting across EtherNet/IP, Profinet, and Modbus TCP, including network setup, addressing, device status checks, and stable data exchange.
- Supported electrical work on the shopfloor by verifying field wiring, terminal mapping, and I/O allocation, fixing wiring mismatches, and coordinating with electrical teams to restore correct signals quickly.
- Worked with AutoCAD electrical drawings by updating wiring diagrams, documenting changes during assembly and commissioning, and keeping drawings aligned with the final built system.
- Built maintenance-friendly systems by adding practical diagnostics, clean manual modes, clear alarm messages, backup and version discipline, and guidance that helped maintenance teams isolate faults faster.
- Handled customer coordination throughout the project, including daily updates during commissioning, clear communication during breakdowns, emergency support for critical production restarts, and technical handover with training and documentation.
- Provided after-sales support through AnyDesk and VPN-based access, using logs and diagnostics to identify root cause, apply safe fixes, and reduce downtime without needing unnecessary site visits.

During the pandemic, my university followed hybrid and online semesters, which made it possible to work and study in parallel. I joined Promatics Solutions in 2021 for full-time, hands-on exposure in industrial automation while completing my B.Tech coursework. With guidance from my faculty, this role was officially recognized as my final-year internship equivalent, and after graduation I continued full-time and supported multiple customer projects through 2024.

Professional Automation Projects

1. Automotive Test Loop Line Automation



A multi-station automotive test loop line where my focus was controlled motion and predictable behavior across interacting stations. I kept the sequence logic clean and the operator flow simple, so faults could be understood fast and recovery stayed safe during real production use.

Overview

- Built step based sequences so conveyors, turntables, presses, and lifts worked in sync.
- Added feedback checks and safety interlocks so motion happened only when conditions were safe.
- Supported commissioning with clear alarms, status, and safe manual controls for quick fault finding.

Client Origin

Germany

Role

PLC and HMI Programmer | Commissioning Support

Technologies Used

Mitsubishi FX5U PLC, GX Works 3, GT Designer 3 (GOT 2000), Ethernet communication, digital IO, sensor feedback handling, interlock logic, Auto and Manual modes, alarm and diagnostics screens, safe motion sequencing.

Project Story

This line was not just one machine. It was a full loop where pallets keep moving from station to station, and everything must stay in sync. Conveyors move pallets forward, turntables rotate them at the right moment, presses perform the action, and lift tables change levels. If one station gets delayed or one sensor behaves unexpectedly, the whole line can stop or create a risky situation.

My responsibility was to make the control logic feel calm and predictable. In Auto mode, the line should move like a well-practiced routine. In Manual mode, the operator should have safe recovery tools that help them fix a problem without creating another one. During commissioning, the goal was always the same: find issues fast, fix them cleanly, and communicate clearly with the customer team.

Objective

Deliver a stable PLC and HMI solution for a multi station automotive test loop line, with safe sequencing, clear interlocks, and operator friendly screens for daily operation and troubleshooting.

System Workflow

- Conveyors: move pallets between stations with sensor verified handshakes.
- Turntable: rotates only when pallet position, stoppers, and station status are confirmed.
- Press station: runs with safety checks and step logic so the cycle is consistent every time.
- Stoppers and lift tables: control spacing and level transitions to avoid collisions.
- HMI layer: shows station status, cycle steps, alarms, and safe manual recovery controls.

How I Worked on Development

- PLC structure and step logic: I built the line as clear steps, so each station knows when to wait, when to act, and when to hand over to the next station. This made testing easier and reduced unexpected behavior during real movement.
- Interlocks and feedback checks: I used real sensor feedback and condition checks before allowing motion. If something was not safe, the logic blocked movement and showed the reason clearly.
- Auto and Manual design: I separated Auto cycles from Manual actions so operators could recover safely. Manual movement was controlled with permissions and clear limits, so small mistakes do not become big incidents.
- HMI usability: I designed screens that answer operator questions quickly: what is running, what is waiting, what is blocked, and what to do next. Alarms were written to be practical, not confusing.
- Communication and diagnostics: I kept Ethernet settings and device communication stable, and added diagnostics bits and status views so faults can be tracked without guessing.

How I Worked on Commissioning and Customer Coordination

- IO checks and validation: I verified sensors and actuators end to end, confirmed correct addressing, and fixed mismatches early so later testing stays smooth.
- Dry run to live trials: I tested sequences step by step, first in safe conditions, then with real movement. When timing issues appeared, I adjusted steps, timeouts, and confirmations until the sequence became reliable.
- Troubleshooting under pressure: when the line stopped, I used the HMI status, station signals, and logic steps to isolate the root cause quickly, then implemented a clean fix instead of a temporary workaround.
- Customer interaction: I coordinated with the customer team during trials, explained what the issue was in simple terms, shared what change I was making, and confirmed the result together. This kept trust high and decisions fast.
- Operator handover: I supported training and explained safe manual recovery flows so the team could handle common issues confidently.

Challenges and What I Fixed

- Sequence breaks due to sensor timing: improved stability using better step confirmation, debouncing where needed, and clear timeouts with alarms.
- Collision risk during recovery: strengthened interlocks and added safer manual permissions so recovery actions stayed controlled.
- Slow fault finding: improved HMI status and alarm clarity so operators and maintenance could locate the problem faster.

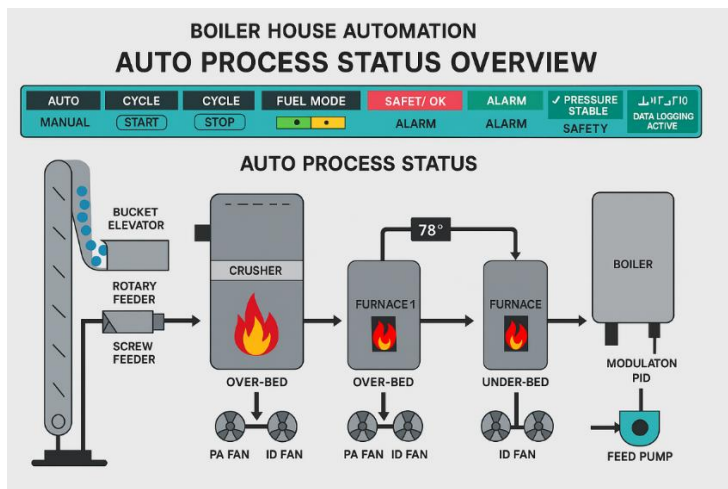
Deliverables and Handover

- Final PLC and HMI project backups with clean structure.
- IO mapping notes and basic station wise signal references.
- Alarm list with messages written for operators.
- Commissioning checklist and trial observations for future support.

Key Outcomes

- Smoother line movement and reliable station handshakes.
- Safer operation through feedback verified interlocks and controlled manual recovery.
- Faster troubleshooting using practical HMI status and alarms.
- Successful customer trials and clear handover with documented backups.

2. Boiler House PLC and SCADA Automation



A boiler house automation project where my focus was stable steam generation performance under real plant conditions. I kept control behavior smooth and visibility clear, so modulation, alarms, and trial running stayed reliable without creating unnecessary complexity.

Overview

- Built safe start-up and shutdown sequences for fuel handling, dual furnaces, draft fans, and feed-water system using permissives and interlocks.
- Implemented steam pressure modulation and feed tank level control using PID loops, stable analog scaling, and clear Auto and Manual behavior.
- Led commissioning with loop checks, cold and hot trials, alarm tuning, and operator handover, including close customer communication during production trials.

Client Origin

India

Role

PLC and SCADA Programmer | Commissioning Engineer

Technologies Used

Siemens S7-1200 PLC, TIA Portal V14, WinCC SCADA (TIA), Ethernet communication, analog input and output scaling, PID control, step sequencing, safety interlocks, trends and data logging, alarm and event handling.

Project Story

This project felt like running a small city of machines. Fuel had to travel through multiple stages, two furnaces had to stay stable, fans had to maintain the right draft, and the feed-water system had to protect the boiler at all times.

On paper, it looks straightforward. On the shop floor, small issues like noisy analog signals or a wrong permissive can stop the whole operation.

My job was to make the system behave like a reliable operator: start only when it is safe, react quickly to abnormal conditions, and recover in a clear, predictable way. At the same time, the SCADA screens had to be simple enough that an operator can understand what is happening in seconds, especially during production trials when everyone is under pressure.

Objective

Deliver a complete boiler house automation with stable steam pressure control, strong safety interlocks, and a WinCC SCADA layer that supports daily operation through clear status, alarms, and trends.

System Workflow

- Fuel handling: elevator, crusher, feeders, and screw conveyors with permissive checks and safe sequencing.
- Combustion: dual furnace operation with controlled feed enable and safety driven stop conditions.
- Air management: FD, PA, and ID fans with feedback checks before allowing fuel feed.
- Feed-water: pumps, tank monitoring, and protection logic to avoid low-water risk.
- Boiler control: steam pressure based modulation with limits, stable tuning, and clear Auto and Manual modes.
- SCADA layer: overview screen, subsystem pages, alarms, setpoints, trends, and logging for troubleshooting.

How I Worked on Development

- PLC structure and sequencing: I broke the process into clean steps so each subsystem knew when to start, when to wait, and when to stop. This reduced random starts and made the logic easy to test during trials.
- Permissives and interlocks: I added practical checks like motor healthy, feedback present, and minimum conditions met, so the system does not run equipment blindly. Operators could clearly see why a start command was not accepted.
- Analog scaling and stability: I handled 4 to 20 mA scaling properly, added range checks, and supported calibration. When signals were noisy on site, I adjusted the approach so values became stable and usable for control.
- PID modulation: steam pressure modulation and feed tank level control were built with safe boundaries. I ensured manual override is smooth and returning to Auto does not create sudden jumps.
- WinCC screens and usability: I designed screens like an operator would use them: clear status colors, direct navigation, visible setpoints, and trends placed where they help during troubleshooting.

How I Worked on Commissioning

- Loop checks and IO validation: I verified every signal end to end, corrected wiring mismatches, and confirmed feedback logic, so the PLC decisions matched the real equipment behavior.
- Cold trials to hot trials: I first validated logic without load, then moved step by step into hot trials, watching how the system reacts in real conditions and fixing issues without risking safety.
- Alarm tuning and recovery flow: I refined alarm messages so they explain the cause and the next action. This helped operators stay calm and recover quickly during production trials.
- Customer communication and coordination: during ramp-up, I coordinated closely with the customer team, clarified what changes were safe, and aligned testing steps with their production schedule. This kept decisions fast and avoided confusion on the floor.

Challenges and What I Fixed

- Steam pressure overshoot and frequent trips: I stabilized it by tuning PID parameters, correcting scaling where needed, and aligning cut-off thresholds with actual process behavior.

- Inconsistent analog readings: I supported recalibration and adjusted scaling and validation, so control and trending became reliable.
- Troubleshooting delays: I improved SCADA visibility with better status mapping and meaningful trends, so the team could find the root cause faster.

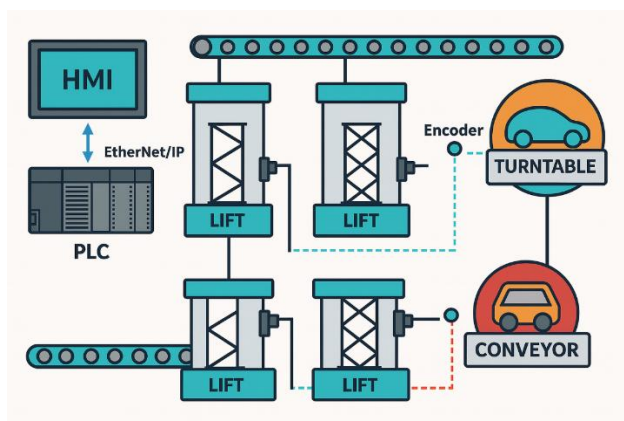
Deliverables and Handover

- Final PLC and WinCC project backups with clean structure.
- IO list, tag list, and scaling parameters for future maintenance.
- Alarm list with clear messages and basic cause and effect notes.
- PID tuning record and recommended setpoint limits.
- Commissioning checklist, trial notes, and handover guidance for operators.

Key Outcomes

- Reliable sequencing and safer operation through strong permissives and interlocks.
- Stable steam pressure modulation and smoother production trials.
- Operator friendly SCADA with alarms and trends that support fast decisions.
- Successful commissioning and handover with documented settings and backups.

3. Multi Lift Conveyor System Integration



A conveyor and lift integration project where my focus was dependable transfers and clean coordination between sections during site acceptance. I validated each interface carefully and aligned testing decisions with the customer team so ramp-up progressed without timing surprises.

Overview

- Integrated new lift stations into an existing conveyor and turntable network without breaking the running sequence.
- Built encoder-based lift positioning with homing, limits, and stable stop accuracy for repeatable vehicle transfer.
- Commissioned on site with tight timelines, coordinated with the customer team, and delivered clean backups plus training.

Client Origin

United Kingdom

Role

PLC and HMI Programmer | On-Site Commissioning Engineer

Technologies Used

Allen-Bradley CompactLogix, Studio 5000, PanelView HMI, FactoryTalk View ME, Ethernet/IP, encoder-based lift positioning, inter-conveyor synchronization, safety relays and interlocks, digital and analog IO, diagnostics and alarm handling.

Project Story

This project was like adding a new lane to a busy highway while traffic is still moving. The customer already had an operating conveyor network, and new lift stations had to be integrated in a way that felt natural to the existing sequence. A lift is not forgiving. If the position is off, the transfer gets risky, cycles break, and everyone loses time.

The goal was not only to make the system run, but to make it behave confidently. Auto mode needed smooth handshakes between conveyors, turntables, and lifts. Manual mode needed safe recovery actions that maintenance can trust. During commissioning, clear communication with the customer team mattered as much as clean logic, because decisions had to be quick and safe.

Objective

Expand and commission a multi-lift and conveyor control system for an automotive facility, coordinating conveyors, lifts, and turntables for continuous vehicle flow between levels, with reliable diagnostics and operator-friendly control.

System Workflow

- Conveyors: transport vehicles between stations using handshake signals and feedback confirmation.
- Lift Units: move vehicles between levels with encoder feedback, homing routines, limits, and position windows.
- Turntables: align vehicles to the next conveyor path with safe permissives and confirmation signals.
- Routing and status: exchange of vehicle or job information and clear status feedback for operations.
- Auto and Manual: full sequence running in Auto, safe maintenance actions and controlled jogging in Manual.

How I Worked on Development

- PLC sequencing and integration: I modified the existing sequence so the new lift stations became part of the natural flow. I focused on predictable state transitions, clean step logic, and clear wait conditions so the system does not rush into unsafe motion.
- Handshake logic between machines: I implemented stable request, acknowledge, and complete signals between conveyors, lifts, and turntables. When timing issues appeared, I adjusted confirmations, timeouts, and sequence conditions so the transfer stayed reliable.
- Encoder-based positioning: I implemented lift positioning using encoder counts with homing, direction checks, limit supervision, and stop windows. This improved repeatability and reduced misalignment during transfers.
- HMI usability and diagnostics: I created and refined screens for Auto and Manual control, interlock visibility, and fault diagnosis. The HMI was designed to answer operator questions fast: what is blocked, why it is blocked, and what is the safe next action.
- Safety thinking: I ensured the logic respects safety chains, permissives, and recovery rules, so manual actions do not bypass critical conditions.

How I Worked on Commissioning and Customer Coordination

- On-site validation: I validated sensors, encoder feedback, and station handshakes on the real equipment and corrected mismatches quickly.
- Trials under real conditions: I ran Auto and Manual trials, observed cycle timing, adjusted sequences, and verified safe recovery flows so maintenance could handle common issues without stress.

- Customer interaction: I coordinated directly with the customer team during the commissioning window, explained root causes in simple terms, aligned on safe changes, and confirmed fixes together.
- Training and handover: I delivered backups and walked the maintenance team through diagnostics, manual recovery, and typical alarms.

Challenges and What I Fixed

- Communication latency between sections: improved handshake stability using better confirmation logic and tuned timing.
- Encoder noise and unstable feedback: handled using filtering and logical validation to avoid false position triggers.
- Tight commissioning window: focused on modular changes, structured testing, and fast coordination to complete validation efficiently.

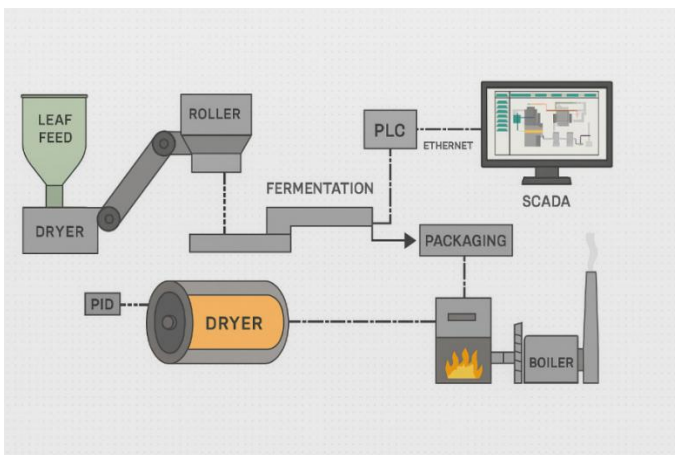
Deliverables and Handover

- Final PLC and HMI backups (Studio 5000 and FactoryTalk View ME).
- Updated IO list, interlock matrix, and encoder related notes.
- Commissioning validation checklist and trial observations.
- Operator guidance for sequence flow, alarms, and safe recovery.

Key Outcomes

- Successful lift integration into an existing conveyor network with stable Auto operation.
- Repeatable lift positioning using encoder feedback and validated stop windows.
- Improved troubleshooting with clearer HMI diagnostics and alarm guidance.
- Completed on-site validation and handover with customer coordination and training.

4. Tea Production Line Process Automation



A tea production line automation project where my focus was steady flow across feeding, drying, fermentation, blending, and packing. I kept the approach modular and practical, so process adjustments remained understandable and day-to-day operation stayed consistent.

Overview

- Built modular PLC logic for conveyors, feeders, dryer, fermentation control, mixer, and packaging sequencing with clear Auto and Manual rules.

- Configured process control for temperature, humidity, and airflow using PID loops, stable analog scaling, and practical limits that match real production.
- Designed WinCC screens, alarms, trends, and daily reporting, then supported commissioning with calibration, trial runs, and operator training plus customer coordination.

Client Origin

India

Role

PLC and SCADA Engineer | Commissioning Support

Technologies Used

Siemens S7 1200 PLC, TIA Portal V16, WinCC SCADA Runtime Advanced, Ethernet communication, analog and digital IO, PID control for temperature and humidity, conveyor and feeder sequencing, dryer and mixer control, safety interlocks, alarm handling, trends, data logging, and production reporting.

Project Story

A tea plant is a chain, not a single machine. If the leaf feed is uneven, the dryer struggles. If drying is unstable, fermentation becomes inconsistent. If blending timing is off, packaging receives the wrong flow. The challenge was to make every section feel connected, so the plant runs like one system instead of isolated islands.

I focused on two things at the same time. First, the PLC logic had to be reliable and safe, with predictable sequences and good interlocks. Second, the SCADA screens had to be practical for operators. When something stops, the operator should immediately see what is blocked, why it is blocked, and what the safe next step is. During commissioning, clear communication with the customer team mattered a lot, because real production always reveals real constraints.

Objective

Develop and commission a complete PLC and SCADA solution for a tea production plant covering leaf feeding, drying, fermentation, blending, and packaging, with stable process control, safe operation, and strong real time visibility through WinCC.

System Workflow

- Leaf feed and conveyors: steady material flow using permissives, sensor feedback, and controlled start rules.
- Drying section: temperature and airflow control using PID loops, limits, and alarm supervision.
- Fermentation and mixing: time based sequencing with environment supervision such as humidity and temperature.
- Blending and packaging: coordinated sequence for blending, weighing, and packaging transfer.
- SCADA layer: overview pages, section pages, alarms, trends, setpoints, and reporting for operations and maintenance.

How I Worked on Development

- PLC structure and modular logic: I built reusable blocks for motors, valves, and section control so the code stayed clean. This helped during testing because each section could be validated without guessing.
- Sequencing and synchronization: I created clear step logic so feeding, drying, fermentation, blending, and packaging stayed coordinated. Handovers between sections were designed with confirmations, not only timers.
- Analog scaling and validation: I handled 4 to 20 mA scaling, range checks, and stable readings so PID control behaved properly.
- PID tuning and limits: I tuned temperature and humidity behavior with safe boundaries, so changes did not shock the process. Manual override and return to Auto were kept smooth to avoid sudden jumps.

- WinCC usability: I built screens that show what matters first, added trends where they help troubleshooting, and wrote alarms that support action, not confusion.
- Reporting and data: I implemented logging and simple production reports so the customer could review history and spot patterns.

How I Worked on Commissioning and Customer Coordination

- Loop checks and calibration: I validated sensors and actuators end to end, supported calibration, and corrected scaling mismatches early.
- Trial runs and stabilization: I ran section wise trials and then full line trials, observing real behavior and adjusting logic and tuning until the plant became stable.
- Customer communication: I coordinated with operators and the customer team during trials, explained what changed and why, and confirmed results together so decisions stayed fast and safe.
- Operator handover: I supported training on alarms, trends, setpoints, and safe recovery steps, so daily operation stayed confident after handover.

Challenges and What I Fixed

- Section synchronization issues: improved handover conditions and confirmations so upstream and downstream stayed aligned.
- Unstable process control: corrected scaling and tuned PID parameters to reduce oscillations during real production.
- Slow fault finding: improved alarm clarity and trend visibility so operators could locate causes faster.

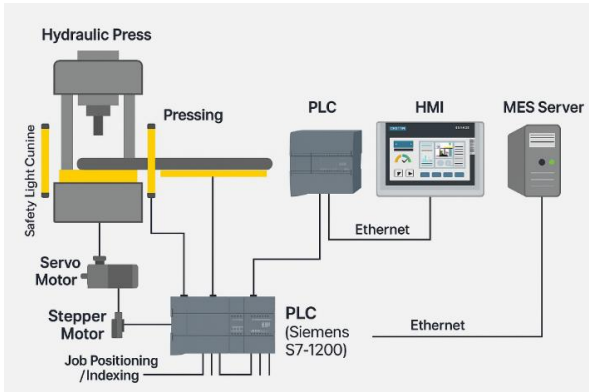
Deliverables and Handover

- Final PLC and WinCC backups with clean structure.
- IO list, tag mapping, scaling parameters, and network notes.
- Alarm list with messages and basic cause and effect notes.
- PID tuning record and recommended setpoint limits.
- Commissioning checklist, trial notes, and reporting templates.

Key Outcomes

- Stable Auto and Manual operation across the full plant workflow.
- More consistent drying and fermentation behavior through improved control.
- Better visibility for operators using clearer alarms and meaningful trends.
- Smooth commissioning handover with documentation and final backups

5. Press Machines Automation with MES



A press automation project where my focus was repeatable cycle behavior, predictable motion performance, and traceability readiness. I also added maintenance-friendly diagnostics and recovery guidance, so service teams could isolate issues faster during acceptance and later support.

Overview

- Developed press sequencing with safe interlocks, part presence checks, and clean Auto and Manual recovery for two different machine workflows.
- Integrated servo and stepper positioning with homing, position windows, timeouts, and repeatable cycle behavior.
- Connected PLC data to MES for traceability and production status, and strengthened maintenance readiness using diagnostics, service-friendly alarms, and guided recovery steps during commissioning.

Client Origin

United States

Role

PLC, HMI and Electrical Design Engineer | Commissioning Support

Technologies Used

Siemens S7-1200 PLC, Siemens KTP700 HMI, TIA Portal V16, Ethernet communication, MES integration, servo motor drive, stepper motor positioning, digital and analog IO, safety relays and interlocks, AutoCAD electrical design, diagnostics and alarm handling.

Project Story

These machines were built for real production, not for demo. One machine handled wheel hub seal and bearing stud pressing, and the second was a hydraulic station press. Press machines look simple from outside, but inside the cycle has many small decisions: when it is safe to clamp, when to allow motion, how to confirm a part is present, how to react if a sensor is late, and how to recover without damaging the tool or the component.

My focus was to make the machine behave like an experienced operator. In Auto mode, every step should feel predictable and repeatable. In Manual mode, maintenance should be able to recover safely with clear permissions. On top of that, the customer wanted production visibility, so the PLC data had to be structured properly for MES communication. During trials and commissioning, communication with the customer team mattered a lot, because small changes in the process can affect quality, cycle time, and safety.

Objective

Design, program, and commission two press automation machines with stable sequencing, reliable servo and stepper positioning, and MES connectivity for production monitoring and traceability, supported by clean electrical documentation and handover.

System Workflow

- Cycle preparation: safety checks, air and hydraulic readiness, part present confirmations, and station permissives.
- Press sequence: clamp, position, press action, confirmation signals, and safe return steps.
- Motion behavior: homing routines, axis enable conditions, position windows, and fault handling for servo and stepper movement.
- MES exchange: structured PLC data blocks for cycle status, result flags, and production events.
- HMI layer: Auto and Manual controls, alarms, diagnostics, and guided recovery steps for operators.

How I Worked on Development

- PLC sequencing and state control: I implemented step based logic so the cycle stays deterministic. Every step had clear entry conditions, confirmations, and timeouts, which made troubleshooting faster during trials.
- Safety and interlocks: I added practical permissives and interlocks around clamping, motion enable, and press actions so the system does not act when conditions are unsafe.
- Servo and stepper positioning: I supported stable positioning behavior using homing, direction checks, limits, and position windows. The goal was repeatability, not only movement.
- MES integration design: I organized PLC data in a clean structure for handshakes and status transfer. This reduced confusion when mapping tags and improved communication reliability.
- HMI usability: I built screens that show machine readiness, current step, interlock blocks, and alarms in a practical way. Operators can see what is stopping the cycle and what action is safe next.
- Maintenance support mindset: I added service-friendly diagnostics, manual jog permissions, and clear alarm hints, then walked maintenance through common recovery scenarios so they can resolve issues quickly during real production.
- Electrical design support: I maintained AutoCAD electrical drawings and kept changes aligned with wiring updates, so the documentation matched the built machine.

How I Worked on Commissioning and Customer Coordination

- IO checks and validation: I validated sensor feedback, actuator behavior, and safety chain responses during trials and corrected mismatches early to avoid later surprises.
- Trial runs and stabilization: I ran cycles step by step, observed timing, tuned timeouts and confirmations, and ensured Auto and Manual behavior stayed consistent.
- Customer communication: I coordinated with the customer team during acceptance activities, explained issues in clear terms, aligned on safe changes, and verified results together.
- Handover and training: I supported operator and maintenance handover with alarms, diagnostics, and recovery guidance, plus final backups and documentation.

Challenges and What I Fixed

- MES handshake reliability: improved consistency by standardizing PLC status bits and using Ethernet diagnostics during testing.
- Servo and stepper synchronization: refined motion behavior using safe acceleration profiles and better confirmation logic to avoid vibration and misalignment.
- Electrical changes during assembly: kept drawings updated and aligned with real wiring so the final documentation stayed accurate.

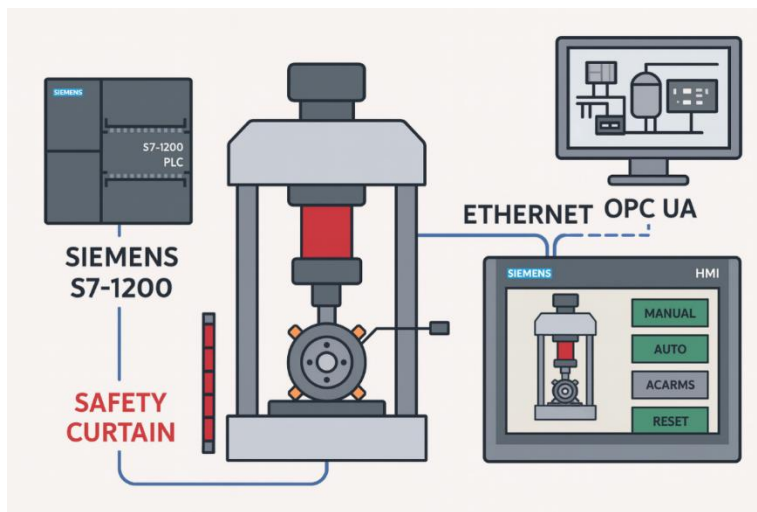
Deliverables and Handover

- Final PLC and HMI backups with MES configuration.
- MES tag mapping and communication reference.
- Servo and stepper parameter sheets.
- AutoCAD electrical drawings and IO mapping tables.
- Alarm list with practical operator messages.
- FAT and SAT checklist and commissioning report.

Key Outcomes

- Stable press cycles with clear interlocks and repeatable motion behavior.
- Better operator confidence through practical HMI diagnostics and recovery steps.
- Reliable MES connectivity for production visibility and traceability.

6. Stud Pressing Line OPC UA Upgrade



A pressing line upgrade project where my focus was structured monitoring integration and long-term service readiness. I verified shopfloor signals and electrical wiring, then ensured data mapping and troubleshooting remained dependable after system went online.

Overview

- Modified PLC and HMI logic to support a new press model, clean Auto and Manual recovery, and maintenance-safe behavior with safety curtain interlocks.
- Implemented OPC UA data mapping for cycle status, alarms, counters, and key sensor feedback so central monitoring could see the real picture in real time.
- Supported commissioning and service readiness through wiring verification, IO validation, operator training, and after-sales troubleshooting with structured diagnostics and clear alarm hints.

Client Origin

Italy

Role

PLC and HMI Programmer | On-Site Commissioning Engineer | After-Sales Support

Technologies Used

Siemens S7-1200 PLC, TIA Portal V14, Siemens KTP HMI, OPC UA communication, Ethernet networking, safety curtain integration, press sequence logic, alarm handling, cycle counters and diagnostics, maintenance-safe manual mode, and wiring and IO validation support.

Project Story

This line was already a working system, but the plant needed two upgrades at the same time. First, the press model was changing, which meant the logic and timings could not stay the same. Second, the customer wanted centralized visibility, so production and fault data had to be shared to the monitoring layer through OPC UA.

The real challenge was doing improvements without creating new confusion for operators and maintenance. So I approached it like a real shopfloor problem. I started by validating what is physically connected, checking sensors, wiring, and signal behavior. Then I updated the PLC and HMI so the sequence stays predictable, safety remains strict, and the data exposed through OPC UA is meaningful, not random bits.

Objective

Modify and validate the PLC and HMI system of an axle and wheel hub stud pressing line, implement OPC UA-based data exchange for centralized monitoring, and deliver a service-friendly system with safe manual operation and clear diagnostics for operators and maintenance.

System Workflow

- Cycle readiness: safety checks, curtain status, press permissives, part presence validation, and station interlocks.
- Press sequence: controlled stroke logic, confirmations, timeouts, and safe return steps.
- Manual and maintenance: safe jog permissions, maintenance blocks, and curtain-based prevention of unintended motion.
- OPC UA layer: status, alarms, counters, and key feedback mapped to a structured node list for monitoring and diagnostics.

How I Worked on Development

- PLC logic updates: I modified the state flow to match the new press behavior, added confirmations and timeouts, and kept Auto and Manual transitions stable so recovery did not feel risky.
- OPC UA data design: I grouped the exposed signals into meaningful sets such as cycle status, alarms, counters, and key sensor feedback. This made monitoring clean and helped remote diagnosis.
- HMI usability: I updated screens to show readiness blocks, safety state, and clear alarm guidance so operators can react quickly.
- Maintenance readiness: I ensured the manual mode is service-friendly with safe permissions, clear interlocks, and diagnostics that reduce guesswork.

How I Worked on Commissioning, Wiring, and Customer Communication

- Electrical wiring verification: I verified sensor placement, checked IO signals end-to-end, and supported quick fixes when wiring or terminal mapping did not match documentation.
- Commissioning validation: I validated both press configurations under Auto and Manual, tested safety curtain behavior, and confirmed stable cycle timing with the customer team.
- Customer communication: I coordinated changes with client engineers and operators, explained why each update was needed, and verified acceptance points together.

After-Sales Customer Support

- Supported troubleshooting after handover using diagnostics and OPC UA monitoring data, helping the team identify whether an issue was wiring, sensor timing, or sequence logic.

- Provided guided recovery steps and minor logic refinements where needed, so downtime stayed low and maintenance confidence stayed high.

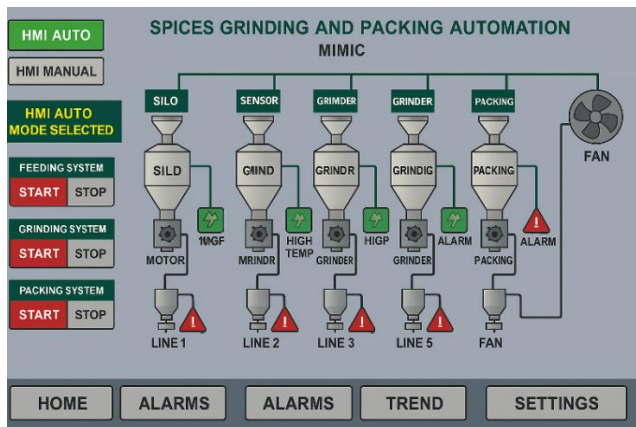
Deliverables and Handover

- Updated PLC and HMI backups aligned with the new press model.
- OPC UA node list and tag mapping reference for monitoring integration.
- Alarm list and practical operator messages for faster fault handling.
- IO and wiring verification notes supporting maintenance reference.
- Commissioning checklist and acceptance validation notes.

Key Outcomes

- Stable Auto and Manual operation with safe maintenance behavior.
- Reliable OPC UA visibility for centralized monitoring and diagnostics.
- Faster troubleshooting through clearer alarms and structured data.
- Better long-term support through after-sales service readiness.

7. Spice Grinding and Packing SCADA System



A multi-line spices plant project where my focus was fast stabilization during an emergency breakdown situation. I verified real field signals and supported electrical wiring corrections where needed, while coordinating with the customer team to restart production safely.

Overview

- Built modular PLC logic for five independent lines so start, stop, interlocks, and alarms stayed consistent across chilly, turmeric, mixed spices, and niche lines.
- Designed line HMIs and a single centralized SCADA view that brings all PLC data and alarms into one operator-friendly dashboard.
- Led time-critical commissioning and recovery during a breakdown: IO validation, alarm linking, VFD checks, emergency electrical wiring fixes, and clear customer communication to restart production safely.

Client Origin

India

Role

Automation and Commissioning Engineer

Technologies Used

Siemens S7-1500 PLC, TIA Portal V14, Siemens HMI, WinCC Runtime Advanced (TIA Portal SCADA), VFD integration, level sensors, temperature transmitters, pressure transmitters, Ethernet communication, alarm management, IO testing, field wiring validation.

Project Story

On paper, this was a clean multi-line project: five separate spice lines, each with its own PLC and HMI, all visible on one SCADA runtime. In real life, it turned into an urgent production recovery. The existing system broke down, and the plant needed production back quickly. That meant no slow debugging, no guesswork, and no risky shortcuts.

I worked like a plant engineer on the floor: first confirm what is actually happening in the field, then fix the real root cause. Some issues were logic-related, some were signal-related, and some were purely electrical wiring mismatches. Once the basics were correct, the SCADA became a powerful tool because operators could see all five lines clearly and react faster.

Objective

Design, test, and commission an integrated automation system for five spice grinding and packing lines, ensuring uniform logic structure, centralized SCADA monitoring, reliable alarms, and fast recovery under emergency production requirements.

System Workflow

- Raw feeding and grinding: material flow with interlocks, motor feedback, and safe start sequencing.
- Blending and packing: coordinated steps with confirmations, timeouts, and operator-ready manual recovery.
- Line instrumentation: level sensors and temperature and pressure transmitters for monitoring and control decisions.
- VFD integration: stable run control, fault handling, and feedback checks for fans and key drives.
- Central SCADA: one place to monitor all lines, acknowledge alarms, view statuses, and support troubleshooting.

How I Worked on Development

- PLC programming for five lines: I built a consistent structure so every line behaved the same way in Auto and Manual. This reduced operator confusion because the logic style was uniform across the plant.
- HMI and SCADA design: I created screens that show readiness, interlocks, running status, alarms, and key signals. I also centralized all five PLCs into one SCADA runtime so the plant had a single monitoring station.
- Alarm strategy and diagnostics: I validated alarms and tag linking end to end so alarms were meaningful and not noise, and the operator could see the cause and the safe next step.

How I Worked on Commissioning, Communication, and Electrical Wiring

- No-load and on-load trials: I tested sequences, safety interlocks, and synchronization first without load, then with real process conditions, adjusting timeouts and confirmations until the cycle became stable.
- Field signal validation: I performed IO testing and verified motor feedback, sensors, transmitters, and VFD status signals so PLC decisions matched real equipment behavior.
- Electrical wiring fixes under pressure: during the breakdown recovery, I supported emergency wiring modifications and field fault corrections, then revalidated signals to ensure the system was safe before restarting production.
- Customer interaction: I coordinated closely with the client team during the emergency window, explained what I was changing and why, and verified acceptance points together so production could restart confidently.

Challenges and What I Fixed

- Central SCADA communication across five PLCs: stabilized using consistent tag mapping and sensible update handling.

- Fan tripping issues: diagnosed with motor protection circuit breaker checks and drive feedback verification, then corrected settings and logic assumptions.
- Emergency downtime risk: reduced through fast root-cause isolation, wiring corrections, and immediate re-commissioning under time pressure.

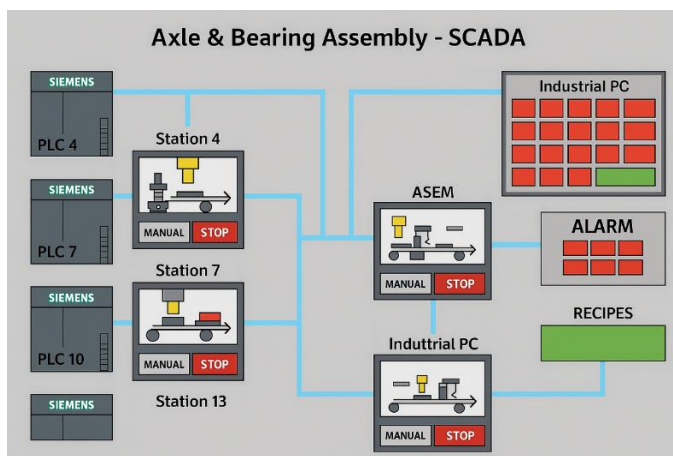
Deliverables and Handover

- Complete SCADA runtime package on the client PC with clear startup flow.
- PLC and HMI backups for all five lines with tag lists and IO allocation sheets.
- Alarm list validation notes and operator guidance for acknowledgment and recovery.
- Commissioning checklist and emergency recovery notes for future maintenance reference.
- Operator training support and documentation handover for daily operation.

Key Outcomes

- All five grinding and packing systems commissioned and monitored under one SCADA station.
- Improved visibility and faster troubleshooting across the plant.
- Successful emergency recovery with wiring corrections and stable production-ready logic.
- Stronger operator confidence due to clearer alarms, screens, and recovery steps.

8. Multi Station Assembly SCADA Optimization



A multi-station supervisory integration project where my focus was usability and maintainability of alarms, tags, and recipe-driven changeovers. I kept configuration structured and consistent, so the system stayed easier to operate and extend across stations over time.

Overview

- Rebuilt and organized a large alarm database of around 1800 alarms using CSV export and import, creating clearer hierarchy, better readability, and faster fault finding.
- Cleaned tag databases and synchronization across four stations and the supervisory PC, then added recipe control for faster product changeovers.
- Implemented safe manual operation for mechanical and hydraulic subsystems and validated end to end behavior with the customer team, making the system easier for operators and maintenance to run daily.

Client Origin

Italy

Role

SCADA Development and Integration Engineer

Technologies Used

ASEM Industrial PC (supervisory), ASEM HMI panels (local stations), CSV tag and alarm configuration, OPC communication, Ethernet and EtherNet/IP, hydraulic and pneumatic press control visualization, recipe management, Siemens S7-1200 and S7-1500 PLCs, TIA Portal V17.

Project Story

This line had four stations running real assembly operations: pressing, locking, conveying, leakage checks, air gap checks, and inspection. Each station had its own ASEM HMI connected to a Siemens PLC, and all stations were tied to a central supervisory ASEM PC. On the surface it looked complete, but in practice the SCADA layer was hard to live with. The alarm database was huge, tags were inconsistent, and troubleshooting took too long because the information was not structured.

I treated it like a real shopfloor problem: the SCADA should help people, not fight them. I started by understanding how operators actually react during stops, then rebuilt the alarm and tag structure so faults became readable. After that, recipe control made a big difference, because changeovers were no longer a manual guessing game. The final goal was simple: if the line stops, the team should quickly know what happened and what is safe to do next.

Objective

Enhance and integrate a four station axle and bearing assembly SCADA environment, with efficient alarm management, reliable tag linkage across all stations, safe manual operation, and recipe based production control through the ASEM supervisory PC communicating with Siemens PLCs.

System Workflow

- Station operations: pressing, locking, conveying, leakage checks, air gap checks, and inspection across stations 4, 7, 10, and 13.
- Local control: each station operated via its local ASEM HMI with direct PLC interface.
- Central supervision: supervisory ASEM PC provided plant level overview, alarm handling, and recipe coordination.
- Data exchange: real time status and alarm exchange over Ethernet with consistent tag mapping and polling.
- Manual and safety: safe manual controls for hydraulic and mechanical subsystems with validation rules.

How I Worked on Development

- Alarm engineering with CSV workflow: I exported the alarm database, reorganized it into clearer categories, standardized naming, removed duplicates, and re-imported it so the structure stayed consistent and maintainable.
- Tag database cleanup and synchronization: I segregated tags by station, validated addresses and references, and ensured the supervisory PC matched the station HMIs so data stayed reliable.
- Recipe management: I implemented recipe logic for both individual stations and synchronized multi station production, reducing changeover time and preventing wrong settings during new product runs.
- HMI usability and diagnostics: I designed animated objects and status views for conveyors, stoppers, presses, and locking mechanisms, so operators could see what is moving, what is waiting, and what is blocked.
- Maintenance readiness: I made alarms more action-focused and added manual controls with safety validation, helping maintenance perform recovery without bypassing critical conditions.

How I Worked on Testing, Integration, and Customer Communication

- Integration validation: I verified communication stability between Siemens PLCs, station HMIs, and the supervisory PC, then tuned polling and consistency checks so updates stayed stable.

- System testing and debugging: I tested alarm triggering, acknowledgments, station readiness states, and recipe transitions, then fixed edge cases that only appear during real operation.
- Customer coordination: I worked closely with Italian automation specialists, explained the reasoning behind structural changes, and aligned acceptance checks so the end result matched how the line was actually used on site.

Challenges and What I Fixed

- No OEM support: reverse engineered SCADA logic and tag references to keep compatibility and stability.
- Very large alarm database: simplified maintenance using a repeatable CSV process and a clean hierarchy.
- Cross station consistency: improved synchronization by validating tags, standardizing naming, and tuning update behavior.

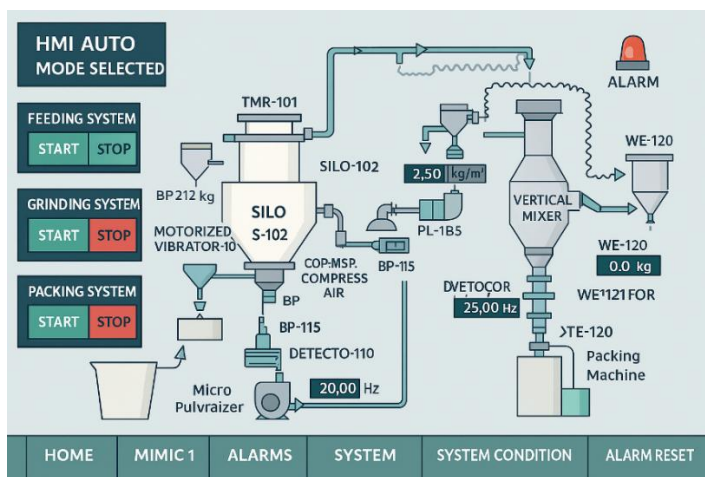
Deliverables and Handover

- Unified alarm and tag list with clear structure and grouping.
- Recipe configuration sheets for both station and full line operation.
- Operator and maintenance guide for alarms, manual operation, and recipe workflow.
- Final project backups and test notes for long term maintenance reference.

Key Outcomes

- Cleaner multi station SCADA integration with reliable synchronization.
- Better diagnostics through structured alarms and clearer screen behavior.
- Faster changeovers using recipe based production modes.
- More confidence for operators and maintenance due to safer manual controls and practical guidance.

9. Pharma Powder Handling Line Commissioning



A pharma powder handling line commissioning project where my focus was stable cycles across conveying, grinding, mixing, and packing. I also strengthened maintenance readiness through practical diagnostics and clear recovery flow, so downtime could be reduced after handover.

Overview

- Validated IO, calibrated analog transmitters, and stabilized Auto cycles for silo feed, grinding, and packing with strong permissives and interlocks.

- Integrated VFD control, setpoints, feedback, and fault handling, then built HMI screens that show what is blocked, why it is blocked, and how to recover safely.
- Commissioned on the shopfloor with customer coordination, added maintenance-friendly bypass and diagnostics, and delivered clear handover documentation plus operator training.

Client Origin

India

Role

Automation and Commissioning Engineer

Technologies Used

CompactLogix 5370 (1769-L19ER), Studio 5000 v32, PanelView Plus 7 (PV2711), FactoryTalk View ME, EtherNet/IP, VFD integration, weight, DP, and level transmitters, metal detector integration, alarm handling, interlocks, and maintenance bypass.

Project Story

This project was all about trust. In a pharma environment, powder handling has no space for uncontrolled starts or unclear alarms. The system moves material from silo to micro-pulverizer, controls feeding through a screw feeder, mixes in a vertical or nauta-style mixer, then goes into weighment and packing. Every section can run independently in Manual, but Auto cycles must behave like a well-practiced routine.

On site, the work was not only programming. It was validation, calibration, and real troubleshooting. When a DP transmitter drifts, the logic makes the wrong decision. When a VFD fault feedback is not mapped cleanly, operators lose time. My goal was to make the system calm: clear permissives, reliable analog values, meaningful alarms, and recovery steps that maintenance can perform safely.

Objective

Commission and validate an automated line for powder conveying, grinding, mixing, weighment, and packing, ensuring reliable Auto and Manual operation, safe interlocks, practical alarm handling, and a smooth handover to operations and maintenance.

System Workflow

- Silo feeding: safe start checks, level confirmations, and controlled conveyance to the grinding section.
- Grinding: micro-pulverizer sequence with screw feeder coordination, VFD behavior, and fault supervision.
- Mixing: vertical or nauta-style mixing sequence with timing, permissives, and safe stop rules.
- Weighment and packing: stable weight readings, pack cycle steps, and clear reject handling when required.
- Safety and quality controls: metal detector reject, door open, DP high, emergency stop, and other plant permissives.

How I Worked on Development

- Auto cycle sequencing: implemented and tested three structured Auto cycles, silo feed, grinding, and packing, with step confirmations and timeouts so the sequence stayed predictable.
- Interlocks and permissives: built safe start logic using DP, level, door status, E-stop chain, and metal detector reject conditions, reducing unsafe starts and false running states.
- VFD integration: configured start and stop, Hz setpoints, feedback mapping, and fault handling so drives responded correctly during real production.
- HMI usability: built mimic views, sub-system controls, alarm banners, and trends, focusing on operator clarity instead of only showing raw tags.

- Maintenance readiness: added controlled bypass logic for specific checks where required, with clear indication and safety validation, so maintenance could diagnose without risky shortcuts.

How I Worked on Commissioning and Customer Interaction

- IO verification and calibration: performed complete IO checks and analog scaling, then supported sensor calibration so transmitters produced stable values.
- Dry runs and simulation: executed dry-run Auto cycles using signal simulation for sections that were not yet wired, which helped validate logic early and reduce surprises later.
- On-floor troubleshooting: isolated root causes using status bits, alarms, VFD diagnostics, and signal checks, then implemented clean fixes rather than temporary workarounds.
- Customer coordination and training: aligned test steps with the customer team, explained changes clearly, and trained operators on alarms, manual recovery, and safe restarts.

Challenges and What I Fixed

- Noisy or drifting analog signals: stabilized by calibration support, scaling validation, and practical range checks in logic.
- Drive fault confusion: improved mapping and HMI messaging so operators could quickly identify fault source and safe next action.
- Unsafe recovery attempts: introduced clearer interlock visibility and safer manual permissions so recovery stayed controlled.

Deliverables and Handover

- Commissioning checklist, parameter sheets, and calibration notes.
- Alarm list with practical messages and recovery hints.
- Operator quick-reference guide for Auto, Manual, and safe restart.
- Final PLC and HMI backups for long-term support.

Key Outcomes

- Stable Auto and Manual operation across silo feed, grinding, and packing.
- Safer start behavior through strong permissives and interlocks.
- Faster troubleshooting using clearer alarms and HMI diagnostics.
- Smoother handover to operations and maintenance with training and documentation.

Education

Master in Digital Engineering

Otto von Guericke University Magdeburg, Germany

04.2024 - Present

Bachelor of Technology in Electrical Engineering

Savitribai Phule Pune University, India

06.2020 - 07.2023

Diploma in Electrical Engineering

Maharashtra State Board of Technical Education (MSBTE), India

07.2018 - 03.2020

Skills

Automation & Control

- PLC Programming (Siemens • Allen-Bradley • Mitsubishi • Beckhoff • Omron • Schneider Electric • Delta Electronics • B&R Industrial Automation • Keyence • Fuji Electric • Phoenix Contact)
- HMI & SCADA Development
- Industrial Commissioning & Troubleshooting
- Safety Interlocks & Control Logic Design
- PID Control & Analog/Digital I/O Configuration
- Motion Control (Servo & Stepper Motors)
- VFD Integration & Drive Commissioning

Engineering Tools

- Siemens TIA Portal (V14–V17)
- WinCC SCADA / Runtime Advanced
- TwinCAT 3 (Beckhoff)
- Studio 5000 (Allen-Bradley)
- GX Works 3 / GT Designer 3 (Mitsubishi)
- Sysmac Studio (Omron)
- EcoStruxure Control Expert / SoMachine (Schneider Electric)
- ISPSOFT / WPLSOFT (Delta Electronics)
- Automation Studio (B&R Industrial Automation)
- KV Studio (Keyence)
- PLCnext Engineer (Phoenix Contact)
- AutoCAD Electrical

Industrial Communication

- Profinet • Profibus • EtherNet/IP • Modbus • OPC UA
- PLC–HMI–SCADA Network Configuration
- MES & Data Exchange Integration

Process Expertise

- Automotive Assembly & Press Lines
- Conveyor & Lift Automation
- Boiler & Utility Automation
- Food / Pharma / Packaging Systems

Key Technologies

- Beckhoff TwinCAT 3 Programming
- Siemens S7-1200 / S7-1500 PLCs
- Safety PLC Configuration (F-Series / TwinSAFE)
- Industry 4.0 & OPC UA Integration

Professional Skills

- On-Site Commissioning
- Cross-Functional Team Collaboration
- Technical Documentation & Operator Training
- Troubleshooting & Root Cause Analysis

Contact Me

Let's Connect

I am always open to discussing new projects, creative ideas, or opportunities to be part of an ambitious team, and I am flexible with location. Feel free to reach out.

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AI Assistant

Ask my AI assistant any questions you have about my experience or skills!

Get In Touch

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